

Chapter 1: Coupled Simulation

In this chapter, we couple the locally mass-conserving MFEM and FVM methods to perform numerical simulations of a partially molten mantle, with the remaining uncertainty in porosity ϕ addressed through phase package evaluation. We describe the implementation details and coupling strategy that integrate the MFEM, FVM, and the eutectic phase package. Before tackling the fully coupled problem, we first examine an alternative porosity evolution scenario based on a prescribed melting rate. We then present an example that combines the MFEM, FVM, and eutectic phase package into a fully coupled computation. The discussion begins with pseudo-1D scenarios and subsequently extends to full 2D simulations.

1.1 Coupled Algorithm

Let $\mathcal{U}^n = (C^n, H^n)$ denote the cell-averaged solution to the transport system (??) at time t^n . Similarly, let $\mathcal{V}^n = (\check{\mathbf{v}}_r^n, \check{\mathbf{v}}_s^n, \check{q}_l^n, \check{q}_s^n)$ represent the solution to the Darcy-Stokes system at the same time t^n . Finally, let $\mathcal{W}^n = (\phi^n, \phi_1^n, \check{T}^n, c_s^n, c_l^n)$ denote the output of the phase calculation at t^n obtained from the transport result $\mathcal{U}^n$.

For each time step the time t^n , we begin by reconstructing the dimensionless mass composition and enthalpy from the cell-averaged solution $\mathcal{U}^n$ using ML-WENO reconstruction at each data point of interest. The eutectic phase package then provides the corresponding phase parameters $\mathcal{W}^n$ based on these point wise values. With $\mathcal{W}^n$ determined in particular with ϕ^n , we solve the Darcy-Stokes system at the fixed time t^n to obtain $\mathcal{V}^n$. Finally, the cell-averaged solution $\mathcal{U}^{n+1}$ at the next time t^{n+1} is then obtained by forward marching of the transport system. The complete procedure is summarized as a coupled solver in Algorithm 1.

Algorithm 1 Sequential Coupled Solver

Let $\mathcal{U}^0$ be given as initial data.

- 1: **for** Time level $n = 0, 1, 2, \dots, n_{\max} - 1$ **do**
- 2: Given $\mathcal{U}^n$, compute $\mathcal{W}^n$ by the eutectic phase package.
- 3: Using ϕ^n in $\mathcal{W}^n$, solve the Darcy-Stokes system (??)–(??) for $\mathcal{V}^n$.
- 4: Given $\mathcal{W}^n$ and $\mathcal{V}^n$, compute dimensionless effective velocity (??) and phase averaged velocity as

$$\check{\mathbf{v}}_{\text{eff}} = \frac{\phi \check{\mathbf{v}}_l + D_i(1 - \phi) \check{\mathbf{v}}_s}{\phi + (1 - \phi)D_i}, \quad \check{\check{\mathbf{v}}} = \phi \check{\mathbf{v}}_r + \check{\mathbf{v}}_s.$$

- 5: Marching forward the transport system (??) to the next time $n + 1$ for $\mathcal{U}^{n+1}$.
 - 6: **end for**
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1.1.1 Discontinuous Porosity

Due to the presence of no melt regions and distinct eutectic phase regions, the reconstructed values of the dimensionless enthalpy and mass concentration may differ on the two sides of an edge shared by adjacent elements. This, in turn, can lead to two different porosity values being computed for the same edge. However, in a conforming MFEM scheme, it is essential to assign a single consistent and physically appropriate porosity value on each element edge.

Therefore, we evaluate the harmonic average of the two values of porosity ϕ_e^+ and ϕ_e^- at each quadrature points on both sides of an element edge e to obtain

$$\bar{\phi}_e = \frac{2\phi_e^+\phi_e^-}{\phi_e^+ + \phi_e^-}. \quad (1.1)$$

The harmonically averaged porosity $\bar{\phi}_e$ is then used in the continuous compaction equation (??), where the face integral is converted to an edge integral via the divergence theorem.

Importantly, when the porosity on one side of the edge vanishes, the geometric average also becomes zero. This property enforces the physical condition that no

liquid can pass directly through an edge adjacent to a no-melt region.

1.2 Pseudo One-Dimensional Column Tests

In this section, we test the coupled algorithm on a two-dimensional $M \times N$ grid with a pseudo one-dimensional compaction column problem. We study a compacting column aligned along the y -direction, defined over the domain $y \in [-H, 0]$ and $x \in [-L, L]$. Two scenarios are investigated.

- Prescribed melting case: The MFEM and FVM solvers are coupled without phase computation. Instead, a simple prescribed melting function $\Gamma(\mathbf{x})$ is used.
- Phase calculation case: Phase transition and porosity evolution are computed simultaneously, providing a complete coupling between the MFEM, FVM, and eutectic phase package.

For the two tests presented below, we fix the domain dimension in x -direction $L = 0.1$. We fix mesh resolution in the x -direction as $M = 2$ and pick different resolutions in y -direction. The domain dimension and mesh resolution in y -direction are chosen accordingly. We fix the parameter $\Theta = 0$ as well. The code is implemented in PETSc/C++, with implementation details given in the appendix.

1.2.1 Prescribed Melting Cases

Before coupling the mechanical, transport and phase packages together into one integrated computation, we first illustrate a simpler model that couples MFEM and FVM solvers without involving the phase package. In this reduced setting, the porosity distribution is not determined by phase-splitting calculations. Instead, we prescribe the melting rate or, equivalently, the porosity evolution through a fixed source function distributed in space as

$$\phi(t) = \Gamma(\mathbf{x}). \tag{1.2}$$

We express conservation of liquid mass by introducing the prescribed melting term $\Gamma(\mathbf{x})$ as a source term on the right-hand side of the advection equation as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_l) = \Gamma, \quad (1.3)$$

where the diffusion term is being dropped for simplicity in this case.

We prescribe the following boundary conditions for the MFEM and the FVM solvers respectively to replicate the behavior of the one-dimensional problem within the two-dimensional grid.

Consider first the Darcy-Stokes MFEM solver. On the bottom edge $y = -H$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with an essential boundary conditions with a constant upwelling velocity v , so

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v, \quad |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu} ds = |e|^{-1} \int_e v ds;$$

- Tangential Stokes velocity is prescribed with an essential boundary condition as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0;$$

- Normal Darcy flux is prescribed with an essential boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} ds = |e|^{-1} \int_e u ds = 0.$$

On the top edge $y = 0$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with natural boundary conditions, i.e., traction free,

$$t_0 = 0;$$

- Tangential Stokes velocity is prescribed with a zero essential boundary condition as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0;$$

- Normal Darcy flux is prescribed with a natural boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} ds = |e|^{-1} \int_e u ds = 0.$$

On the left and right Edges $x = \pm L$, impose that:

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions as

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0, |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu} ds = 0;$$

- Tangential Stokes velocity is prescribed with the natural boundary condition of free traction,

$$t_0 = 0;$$

- Normal Darcy flux is prescribed with a non-permeable essential boundary condition as

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} ds = |e|^{-1} \int_e u ds = 0.$$

Finally, for the transport FVM solver, on the bottom edge $y = -H$, impose that:

- A corresponding Dirichlet boundary condition is prescribed on the inflow boundary aligning with the essential boundary condition specified in the MFEM solver.

On the top edge $y = 0$, impose that:

- A free outflow boundary condition is imposed, consistent with the free traction boundary condition applied in the MFEM solver.

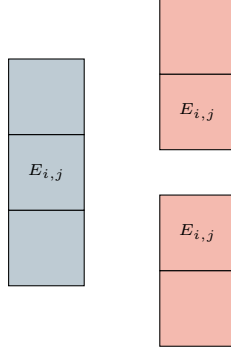


Figure 1.1: An exhibition of a set of 1D ML-WENO (3,2) stencils on a rectangular mesh.

On the left and right edges $x = \pm L$, impose that:

- A no-flow boundary condition is prescribed consistent with the zero-flow Dirichlet boundary used in the MFEM solver.

For simplicity and efficiency in implementation and computation, we employ one-dimensional ML-WENO(3,2) stencils as illustrated in Figure 1.1, since higher-order accuracy in the x -direction is unnecessary for this problem. We use second order SSP Runge-Kutta time integration scheme defined in Equation (??).

We consider two different scenarios, each characterized by distinct prescribed melting functions and initial conditions:

Case 1: A piece-wise constant source function.

$$\Gamma_1(z) = \begin{cases} 2.5e - 3, & -1.5 < z \leq 0, \\ 0, & -2.0 \leq z \leq -1.5. \end{cases} \quad (1.4)$$

We assign an initial condition of no melting everywhere as

$$\phi_1(z, 0) = 0.0, \quad z \in [-2.0, 0.0]. \quad (1.5)$$

Figure 1.2 illustrates the temporal evolution of porosity, Stokes and Darcy velocities, Stokes and Darcy pressures, as well as the effective pressure, defined as $p_{\text{eff}} = p_s - p_l$.

The third panel, corresponding to dimensionless time $\check{t} = 1325$, demonstrates that the system has reached a steady state after sufficient evolution.

Case 2: We next consider a slightly more complicated scenario in which a melting source is trapped beneath the bottom of an existing porosity channel. The trapped melting source produces a step in porosity, thereby induces an instantaneous perturbation in the compaction rate. The source function is

$$\Gamma_2(z) = \begin{cases} 0, & -1.5 < z \leq 0, \\ 1.5e - 3, & -1.8 < z \leq -1.5, \\ 0, & -2.0 \leq z \leq -1.8, \end{cases} \quad (1.6)$$

where we assign an initial condition with an existing porosity channel as

$$\phi_2(z, 0) = \begin{cases} 5e - 2, & -1.5 < z \leq 0, \\ 0, & -2.0 \leq z \leq -1.5. \end{cases} \quad (1.7)$$

Figure 1.3, 1.4 and 1.5 illustrate the temporal evolution of porosity, Stokes and Darcy velocities and Stokes and Darcy pressures. A propagating discontinuity in porosity forms when the porosity profile decreased upward (See Katz (2022)). Eventually, the system reaches a steady state, characterized by a narrower porosity channel. The simulation demonstrates robust and stable evolution, despite the presence of a highly complicated porosity profile. The solution preserves its overall structure under mesh refinement.

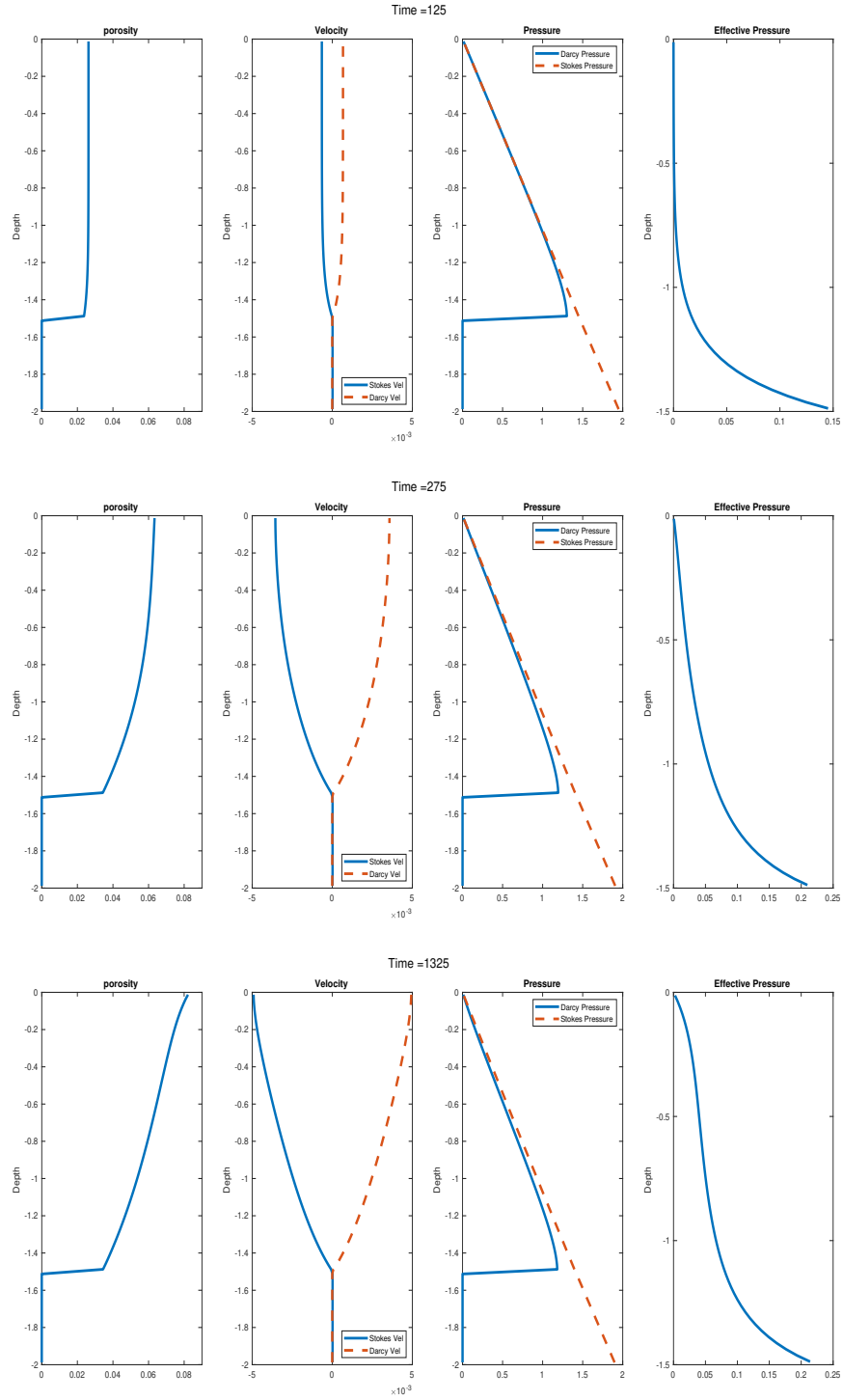


Figure 1.2: Prescribed melting case with constant melting source term 1.4. Shown at time = 125, 275 and 1325, where the system reaches steady state at the time = 1325.

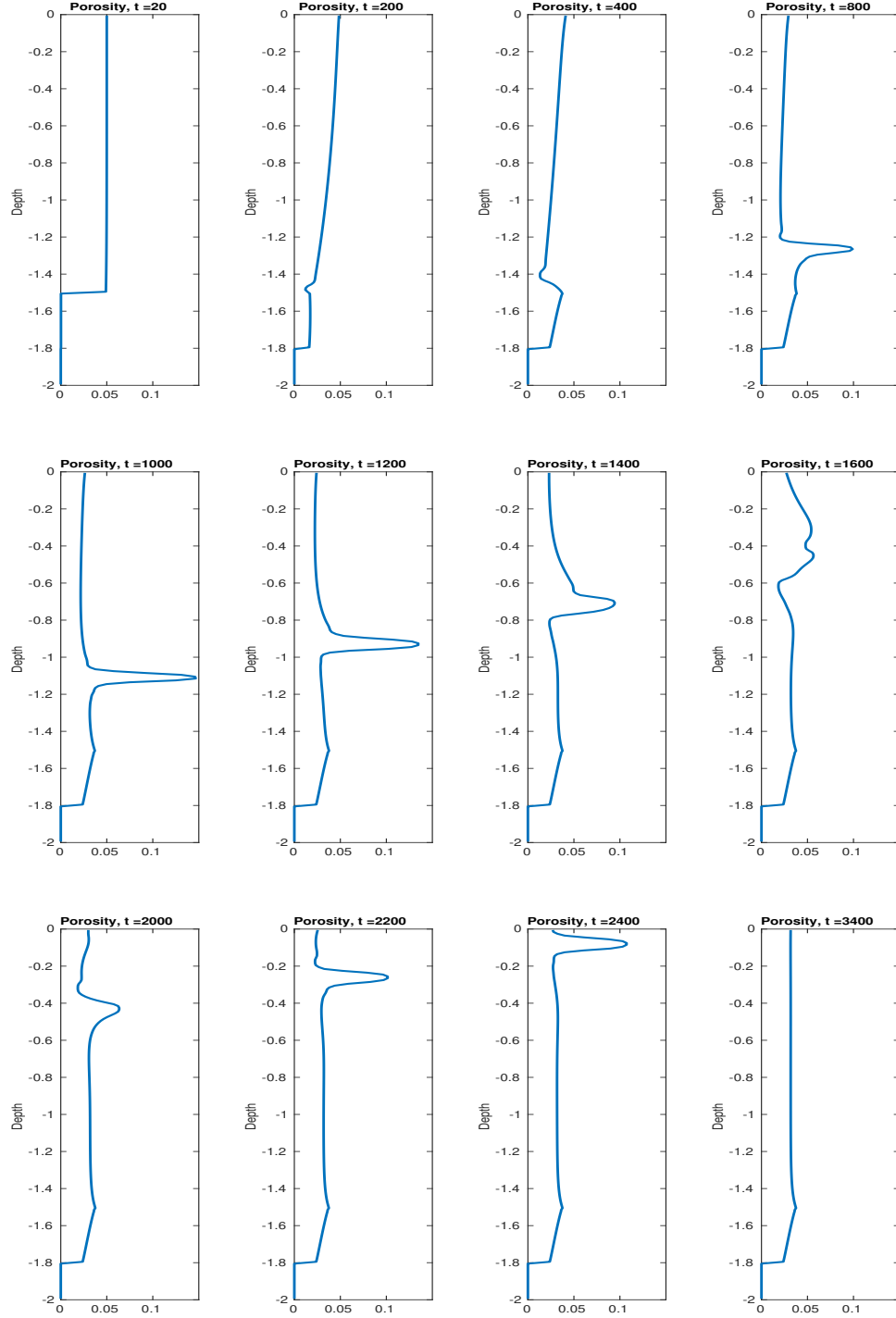


Figure 1.3: Prescribed melting case with trapped melting source term 1.6. Temporal evolution of porosity shown at different time steps.

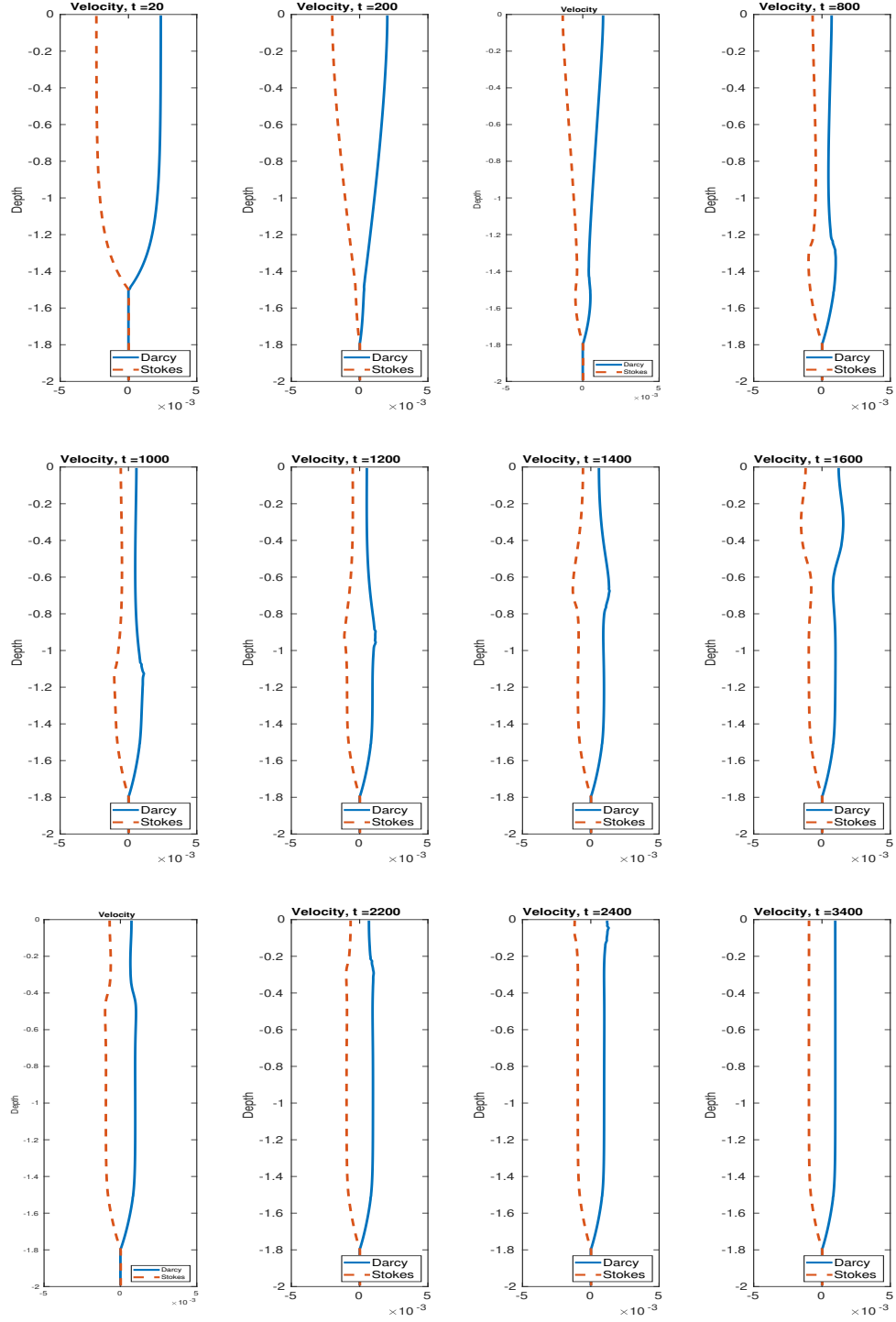


Figure 1.4: Prescribed melting case with trapped melting source term 1.6. Temporal evolution of velocities shown at different time steps.

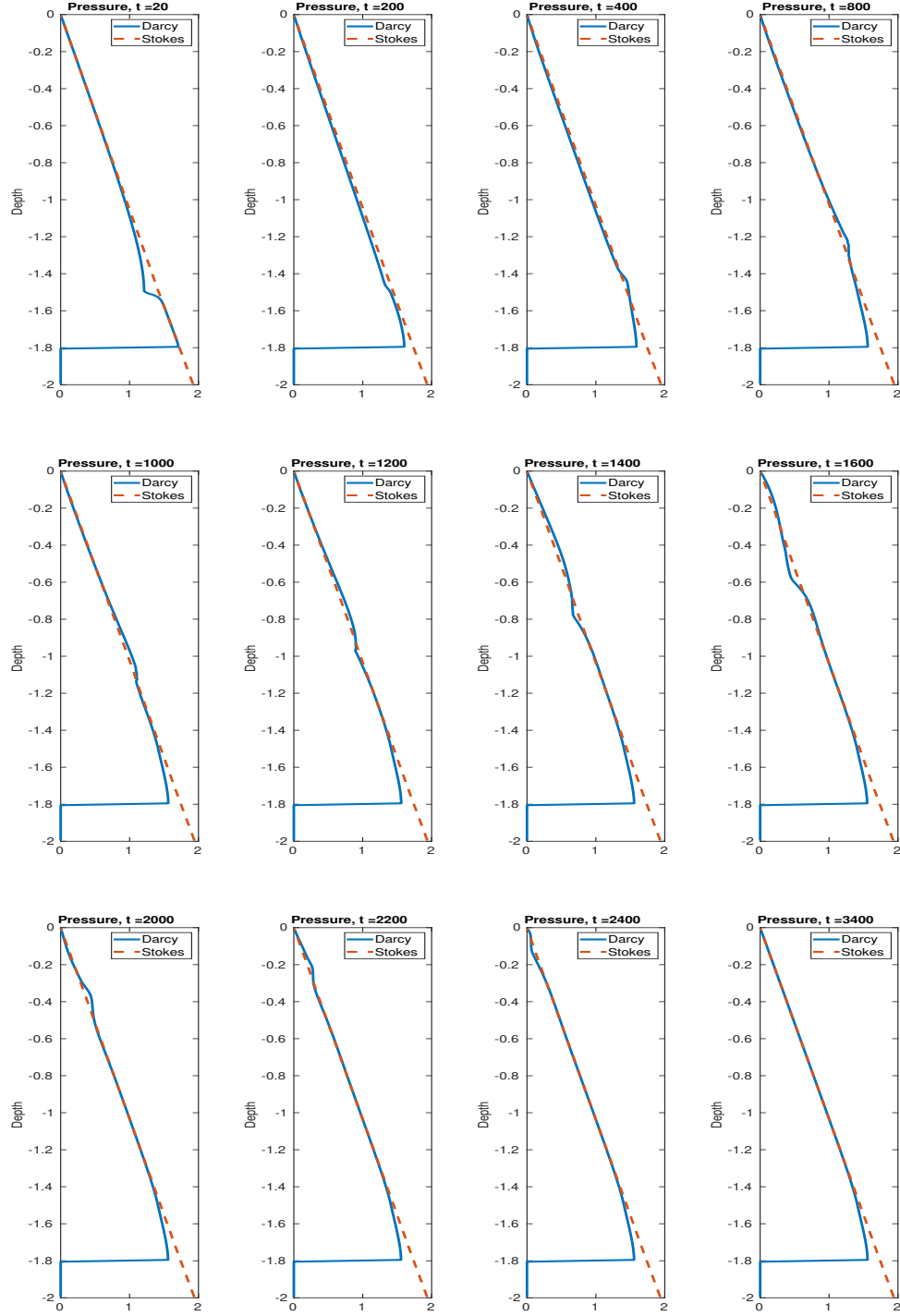


Figure 1.5: Prescribed melting case with trapped melting source term 1.6. Temporal evolution of pressures shown at different time steps.

1.2.2 Phase Coupled Calculation

In this section, we test the MFEM and FVM solvers combined with the eutectic phase package, which gives the fully integrated simulation framework. In this setting, porosity is no longer prescribed through an externally imposed melting function but is instead computed consistently from the transported nondimensional bulk composition and enthalpy.

The computational domain represents an upwelling mantle column, with corner flow neglected near the surface under the simplifying assumption that the column continues its vertical ascent through the lithosphere. This configuration constitutes a minimal yet dynamically representative test case for assessing the stability and accuracy of the coupled algorithm in the presence of phase transitions and melt–solid interactions.

We propose some simplifications for our simulations. First of all, we reconsider the transport equations (??) and the simplified version is

$$\begin{aligned} \frac{\partial C}{\partial \tilde{t}} + \nabla \cdot \check{\mathbf{v}}_{\text{eff}} C &= 0, \\ \frac{\partial H}{\partial \tilde{t}} + \nabla \cdot (\check{\mathbf{v}} \check{T}) - \check{L} \nabla \cdot ((1 - \phi) \check{\mathbf{v}}_s) - \frac{\alpha_\rho l_0}{c_p} \check{T} \check{\nabla} \cdot \mathbf{g} &= 0. \end{aligned} \tag{1.8}$$

In this formulation, the diffusive contributions in both the mass composition and enthalpy equations are neglected. Specifically, the diffusion term $\frac{\phi \mathcal{D}_l}{u_0 l_0} \nabla c_l$ in the liquid phase is omitted due to the fact that the porosity ϕ remains small ($\phi \ll 1$) and the associated diffusivity coefficient $\mathcal{D}_l$ is comparatively negligible compared to effective advective velocities and thus can be neglected (See ?).

Likewise, while thermal conduction is an important mechanism in mantle convection over geological timescales, its influence on the present simulations is negligible. Two considerations motivate the simplification of dropping the thermal conduction from the enthalpy equation (??). First, the temperature profile maintains either a nearly constant or linear variation in space, such that the Laplacian of temperature vanishes or remains small throughout the evolution. Second, we do not impose

a non-permeable solidification boundary at the surface, which would otherwise establish sharp thermal gradients and necessitate conductive heat transport across a boundary layer, in which case the diffusivity would no longer be negligible (see Hewitt and Fowler (2008)).

Additionally, given that the primary focus of this study is the phenomenon of partial melting under small-porosity conditions, we adopt the lithostatic pressure as the reference pressure in determining the local melting point (see Katz (2008)). This approximation is justified because the dynamic pressure perturbations associated with porous flow are small compared to the background lithostatic gradient. As the mantle column ascends from depth, the concomitant reduction in lithostatic pressure leads to a decrease in the solidus temperature, thereby inducing some degrees of melting in proportion to the degree of decompression.

1.2.2.1 Boundary conditions

In this set of simulations, we do not attempt to prescribe fully physically consistent boundary conditions, owing to the inherent complexity of capturing both the mechanical and thermochemical constraints at the mantle–lithosphere interface. For example, Hewitt and Fowler (2008) formulated boundary conditions that simultaneously satisfy conservation laws and thermodynamic equilibrium at the surface. These constraints introduce additional complexity to the model and the numerical scheme that is not central to the phenomena under investigation here.

By prescribing essential boundary conditions on the bottom and the lateral edges and a natural (free-traction) boundary condition at the top for the Darcy–Stokes MFEM solver, the simulations allow for mechanically reasonable mantle upwelling without artificial confinement. Similarly, the Dirichlet inflow and free outflow conditions for the transport solver ensure a constant supply of chemical species and energy while permitting realistic advective transport, without imposing restrictive thermodynamic constraints. This approach maintains the internal consistency of the coupled

MFEM–FVM solver and captures the key dynamics of melt migration, porosity evolution and phase transitions, while reducing computational complexity.

1.2.2.2 Numerical Flux

We modify the semi-discretized formulation of a general conservation law, as presented in equation (1.9), to incorporate a “source” term S . The resulting expression can be written as

$$\frac{\partial \bar{u}_E}{\partial t} + \sum_{i=0}^4 \frac{|e_i|}{|E|} \sum_{g=0}^G \omega_{i,g} F(\mathcal{R}_{i,g}^+, \mathcal{R}_{i,g}^-) + \sum_{h=0}^H \xi_h S(\mathcal{R}_h) = 0, \quad (1.9)$$

where $\omega_{i,g}$ represents the weight associated to the g -th quadrature point on edge e_i and ξ_h represents the weight associated to the h -th quadrature point on the cell. In practice, we employ the midpoint rule for the numerical integration over cells. This allows us to approximate the face contributions using the cell-averaged transport variables while retaining second-order accuracy.

The adiabatic cooling term $\frac{\alpha_{\rho l_0}}{c_p} \check{\tilde{\mathbf{T}}} \cdot \mathbf{g}$ in equation (1.8) is treated as a “source” term, which is straightforward to implement within the semi-discretized formulation. The corresponding Lax-Friedrichs numerical fluxes of the mass composition and the enthalpy across element edges are expressed in terms of the ML-WENO reconstructed values $\mathcal{R}_C$ and $\mathcal{R}_H$ as

$$\begin{aligned} F_C(\mathcal{R}_C^+, \mathcal{R}_C^-) &= \frac{1}{2} \check{\mathbf{v}}_{\text{eff}} \cdot \boldsymbol{\nu} \left[\mathcal{R}_C^+ + \mathcal{R}_C^- - \alpha_C (\mathcal{R}_C^+ - \mathcal{R}_C^-) \right], \\ F_H(\mathcal{R}_H^+, \mathcal{R}_H^-) &= \frac{1}{2} \check{\tilde{\mathbf{v}}} \cdot \boldsymbol{\nu} \left[\check{\tilde{T}}^+ + \check{\tilde{T}}^- - \alpha_T (\check{\tilde{T}}^+ - \check{\tilde{T}}^-) \right] - \check{L}(1 - \phi) \check{\mathbf{v}}_s \cdot \boldsymbol{\nu}, \end{aligned} \quad (1.10)$$

where $\boldsymbol{\nu}$ is the outward unit normal of the edge, and $\check{\mathbf{v}}_{\text{eff}}$ and $\check{\tilde{\mathbf{v}}}$ denote the effective velocity and phase averaged velocity continuous across the edges. The terms $\check{\tilde{T}}^+ = \check{\tilde{T}}(\mathcal{R}_H^+, \mathcal{R}_C^+)$ and $\check{\tilde{T}}^- = \check{\tilde{T}}(\mathcal{R}_H^-, \mathcal{R}_C^-)$ are temperatures evaluated on both sides of the edge using the eutectic phase package, based on the reconstructed pointwise mass composition and enthalpy. The stabilizers α_C and α_T use the local Lax-Friedrichs scheme as shown in equation (??), ensuring stability of the flux discretization.

The calculation of the numerical flux (1.10) of the mass composition is not entirely straightforward. The primary transport variables—mass composition and enthalpy—may undergo distinct reconstructions across an edge due to discontinuities and the presence of multiple phase regions. As a consequence, the porosity ϕ can take different values, denoted $\phi^+ = \phi(\mathcal{R}_H^+, \mathcal{R}_C^-)$ and $\phi^- = \phi(\mathcal{R}_H^-, \mathcal{R}_C^-)$. To ensure compatibility in the MFEM discretization, we adopt the harmonic average of the two values, as discussed previously in Section 1.1.1, and defined the compatible porosity on the edge as

$$\bar{\phi}_e = \frac{2\phi_e^+\phi_e^-}{\phi_e^+ + \phi_e^-}.$$

In contrast, the bulk mass composition represents a relatively smooth advected quantity. A harmonic average would add artificial damping and suppress mixing, while an arithmetic average is more suitable without over-penalizing discontinuities. Therefore, we adopt

$$c^e = (c^+ + c^-)/2. \quad (1.11)$$

We now revisit the definition of effective velocity defined in equation (??) with these interface definitions as

$$\mathbf{v}_{\text{eff}} = \frac{c_l^e \bar{\phi} \mathbf{v}_l + c_s^e (1 - \bar{\phi}) \mathbf{v}_s}{c_l^e \bar{\phi} + c_s^e (1 - \bar{\phi})}, \quad (1.12)$$

where $\bar{\phi}$ is the harmonic-averaged porosity and c_s^e, c_l^e are the arithmetic averaged mass compositions.

Similarly, the phase averaged velocity required in the enthalpy flux is expressed without ambiguity as

$$\check{\mathbf{v}} = \bar{\phi} \mathbf{v}_l + (1 - \bar{\phi}) \mathbf{v}_s, \quad (1.13)$$

where the same harmonic averaged porosity is employed. This guarantees that in the limiting case $\phi^+ = 0$ or $\phi^- = 0$, the phase-averaged velocity reduces to the solid velocity, thereby preserving the correct physical behavior and preventing artificial transport of enthalpy associated with liquid in no-melt regions.

1.2.2.3 Test results

Again, for simplicity and efficiency in implementation and computation, we employ one-dimensional ML-WENO(3,2) stencils as illustrated in Figure 1.1 for both the mass composition and the enthalpy. We use second order SSP Runge-Kutta time integration scheme defined in Equation (??) as well.

Full coupled column testing case $H = 2$: the domain in y -direction spans two compaction length $H = 2$ with mesh resolution $N = 80$. We assign an initial condition of the bulk dimensionless mass composition and the bulk dimensionless enthalpy as

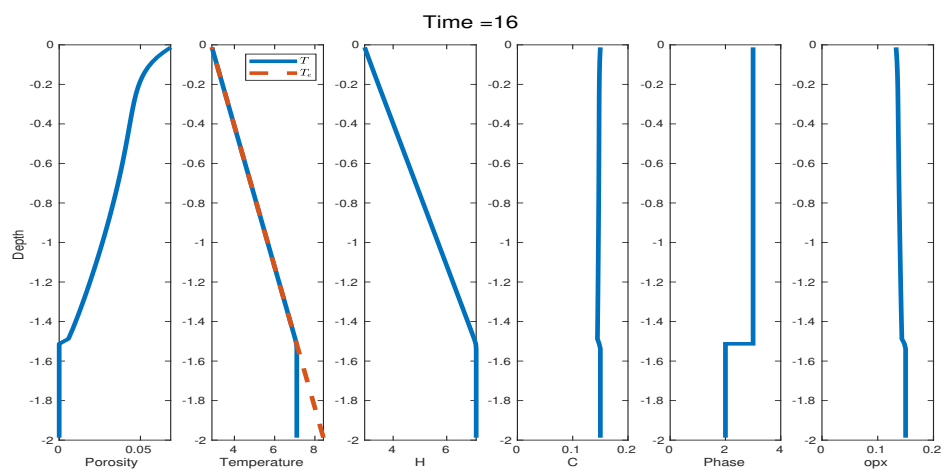
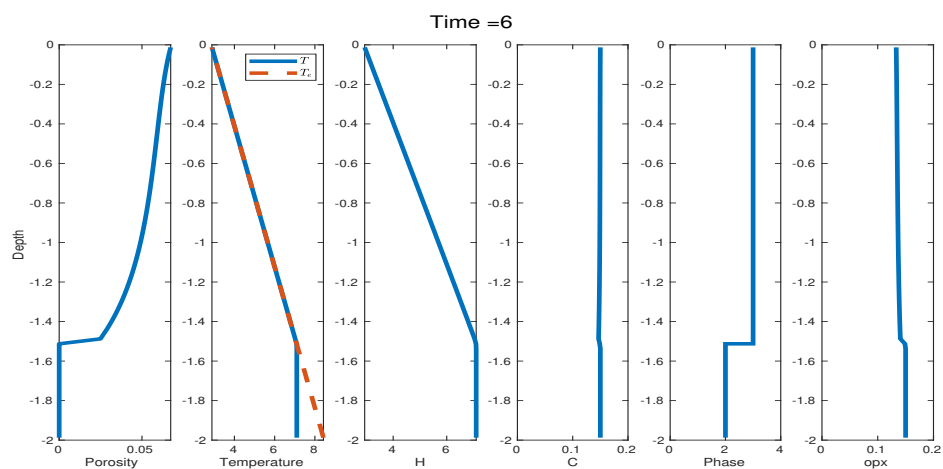
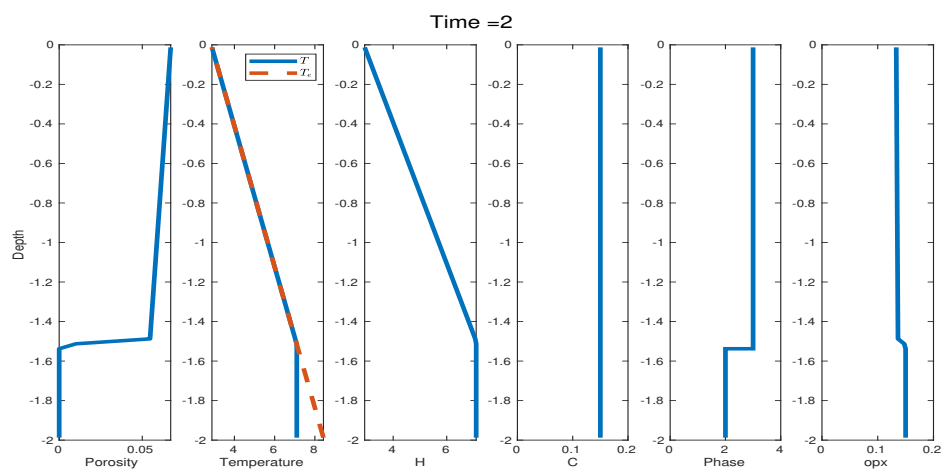
$$H = \begin{cases} 2.9 - 2.8y, & -1.5 \leq y < 0, \\ 2.9 + 2.8 \times 1.5, & -2 \leq y < -1.5. \end{cases} \quad (1.14)$$

$$C = 0.15.$$

Figure 1.6 illustrates the temporal evolution of the primary transport variables, bulk dimensionless mass composition and bulk dimensionless enthalpy, along with the corresponding phase calculation determined from the phase package. From time $t = 2$ to time $t = 6$ we observe a transition from phase region 3 (eutectic melting) to phase region 2 (solid, no melting phase) near the cut $y = -1.5$. The porosity field appears to be highly sensitive to these changes in bulk enthalpy and mass composition, leading to the development of a porosity wave near the upper boundary. This wave-shaped pattern arises from porosity accumulation near boundary, which depletes the mass composition more rapidly than it can be replenished from below. This in turn reduces porosity locally and blocks the draining of liquid further down. This wave-shaped pattern near the top boundary subsequently amplifies and ultimately results in an unstable outcome in the numerical solution around time $t = 120$. The wave-shaped pattern preserves under mesh refinement.

Figure 1.7 shows three characteristic velocities defined on the element edges: the effective velocity transporting chemical species, the phase averaged velocity governing the advection of enthalpy, and the solid velocity carrying latent heat. The

result confirms that the local mass conservation is well-preserved as evidenced by the phase averaged velocity remaining constant under the Boussinesq approximation.



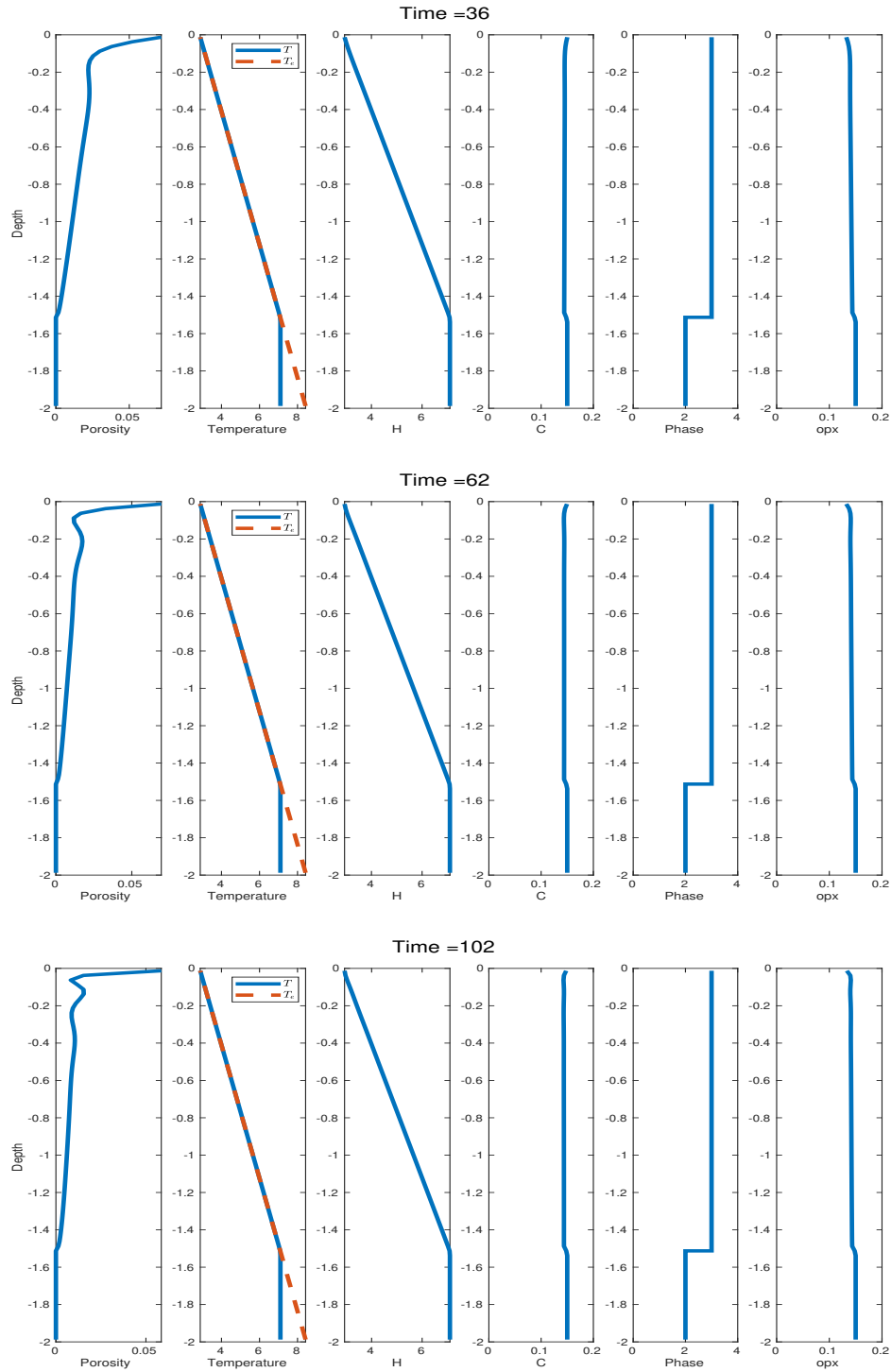
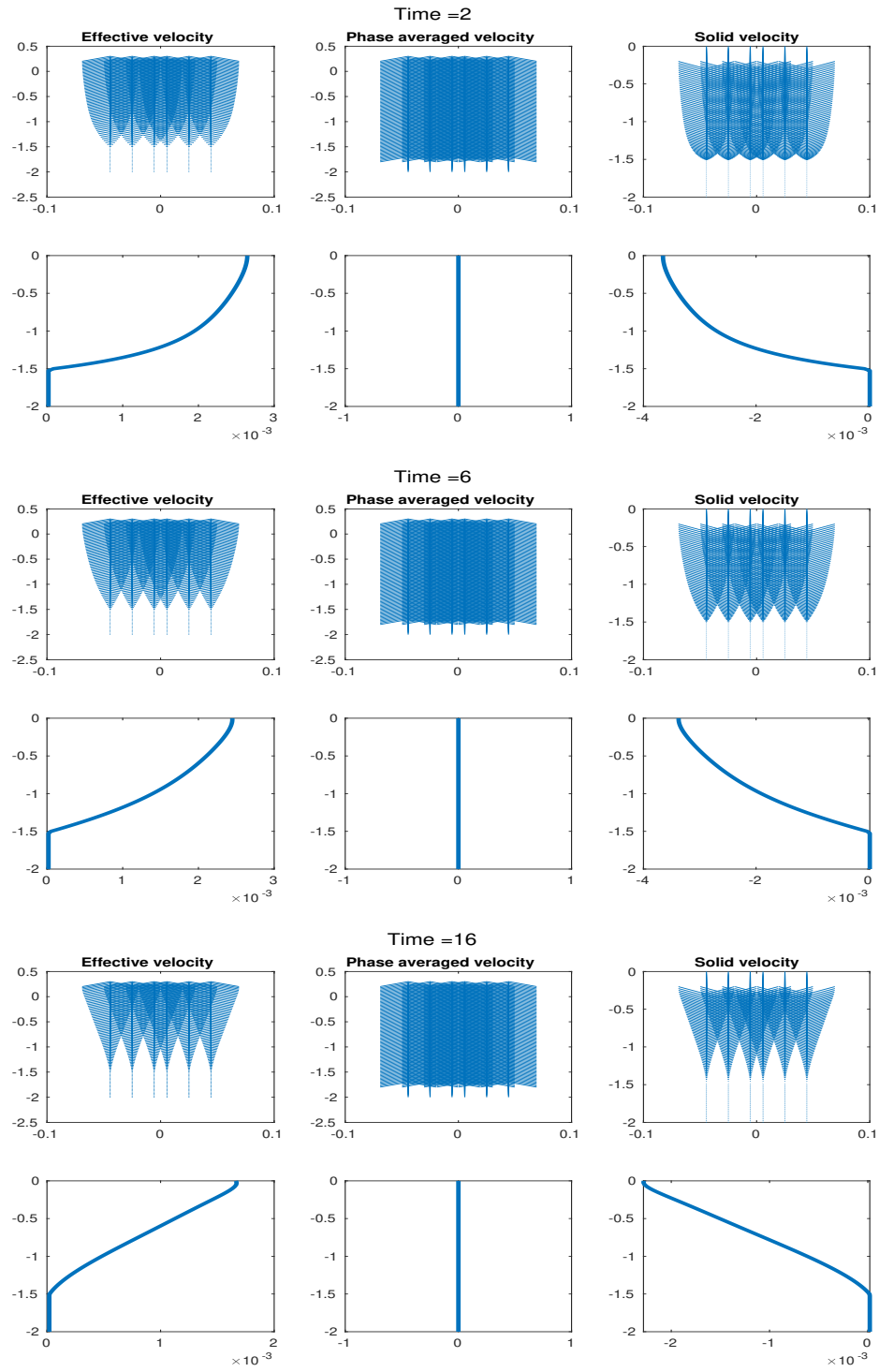


Figure 1.6: Full coupled column testing case $H = 2$. Shown at time = 2, 6, 16, 36, 62 and 102. We present porosity, temperature, dimensionless enthalpy, dimensionless mass composition, phase regions and Opx percentage from left to right.



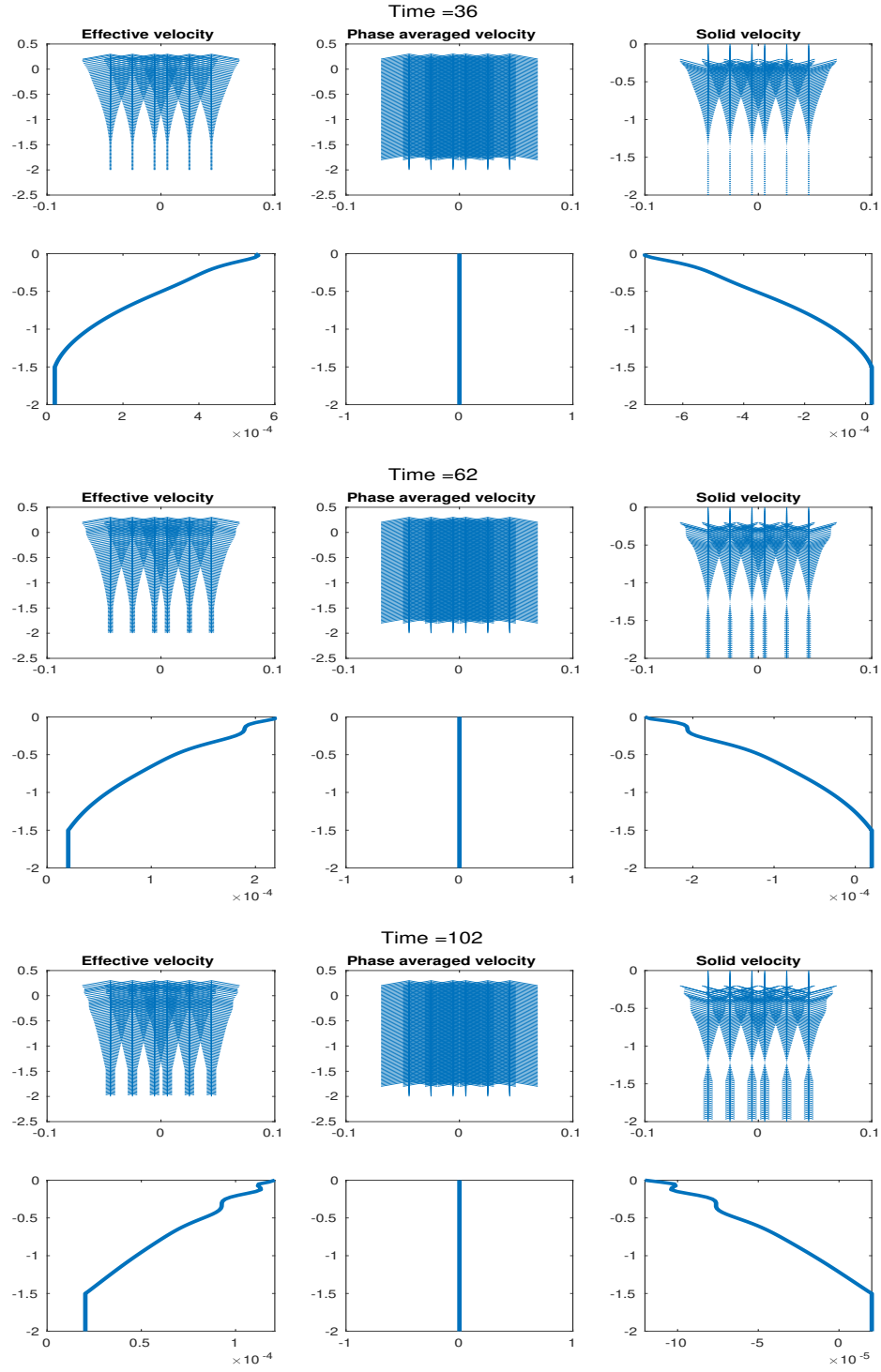


Figure 1.7: Full coupled column testing case $H = 2$. Shown at time = 2, 6, 16, 36, 62 and 102. We present effective velocity, phase averaged velocity and solid velocity from left to right.

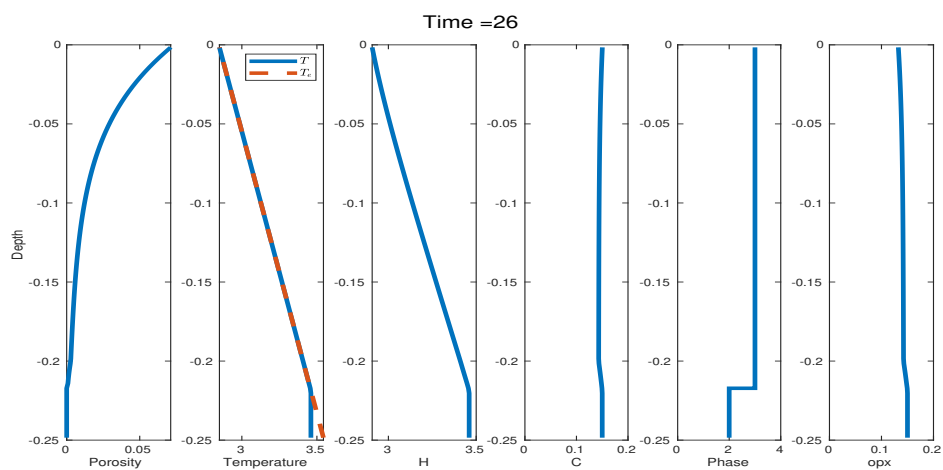
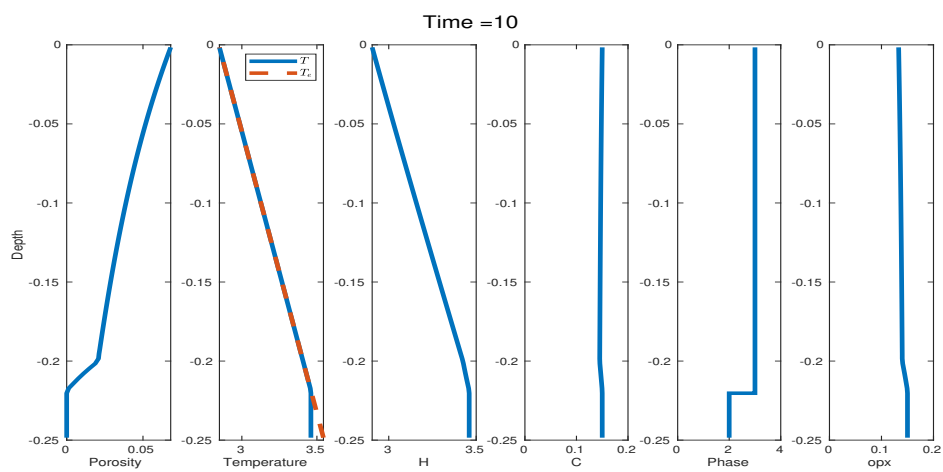
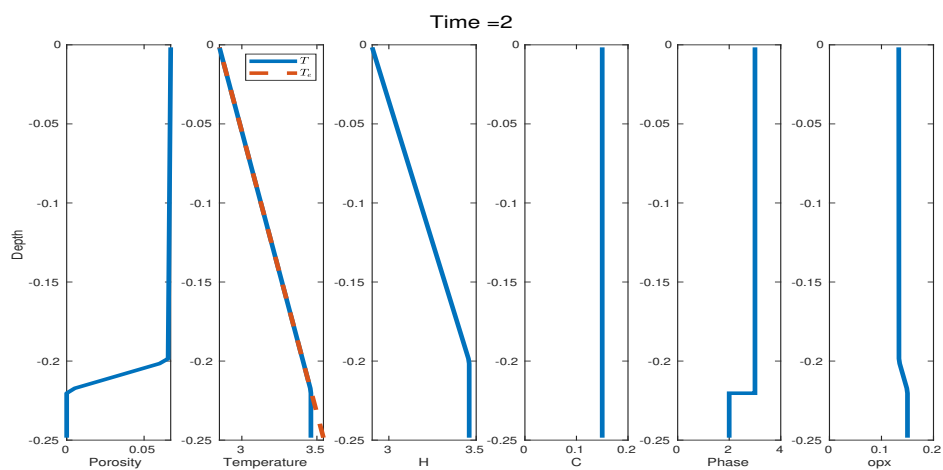
Case 2: the domain in y -direction spans 0.25 compaction length $H = 0.25$ with mesh resolution $N = 80$. We assign the same initial condition of the bulk dimensionless mass composition and a similar initial condition of the bulk dimensionless enthalpy with different cutoff as

$$H = \begin{cases} 2.9 - 2.8y, & -0.2 \leq y < 0, \\ 2.9 + 2.8 \times 0.2, & -0.25 \leq y < -0.2. \end{cases} \quad (1.15)$$

$$C = 0.15 \quad .$$

Figure 1.8 illustrates the temporal evolution of the primary transport variables - bulk dimensionless mass composition and bulk dimensionless enthalpy - along with the corresponding phase calculation determined from the phase package. Between time $t = 10$, time $t = 26$ and time $t = 40$, the system undergoes successive transitions: initially from Phase Region 3 to Phase Region 2, and subsequently returning to Phase Region 3. These transitions occur while maintaining numerical stability throughout the simulation, illustrating the ability of our framework to handle transition between melting and no melting zones. The system reaches steady state at time $t = 240$, while a notable accumulation of porosity remains concentrated near the top boundary.

Figure 1.9 again presents the three characteristic velocities defined on the element edges: the effective velocity transporting chemical species, the phase averaged velocity governing the advection of enthalpy and the solid velocity carrying latent heat. As the simulation progresses, the effective velocity exhibits a marked increase near the top boundary, consistent with the porosity accumulation observed in Figure 1.8.



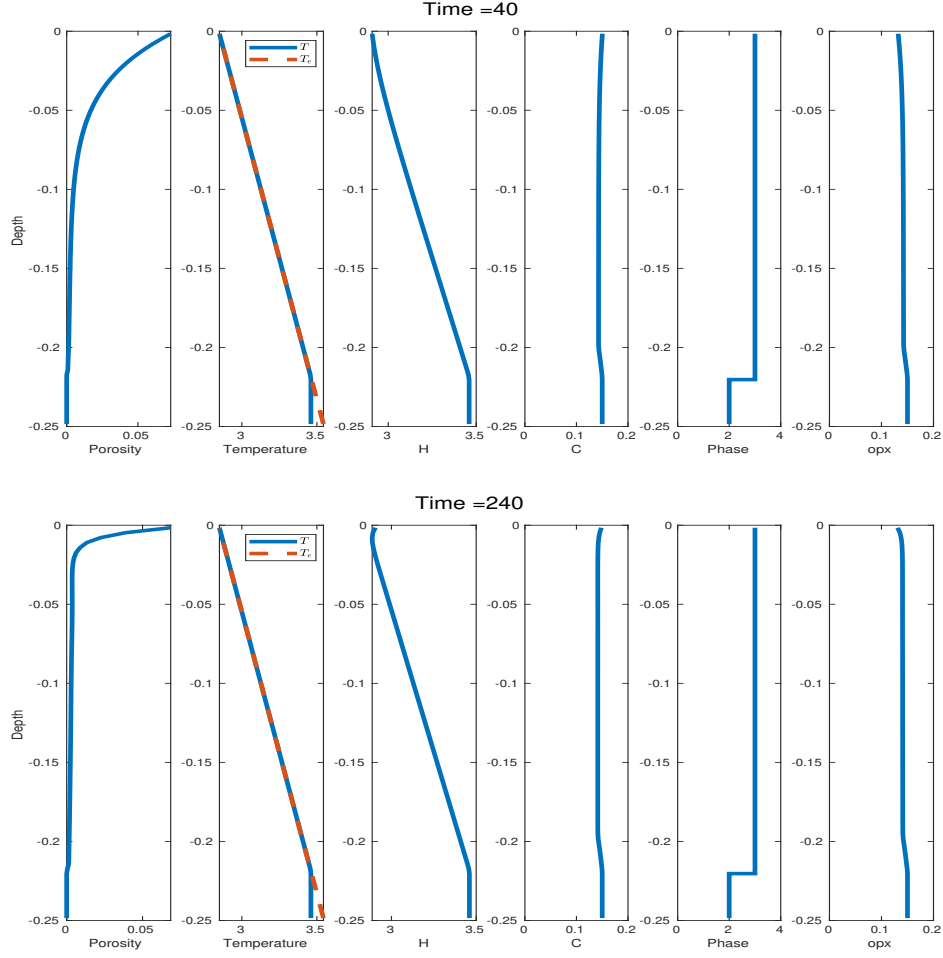
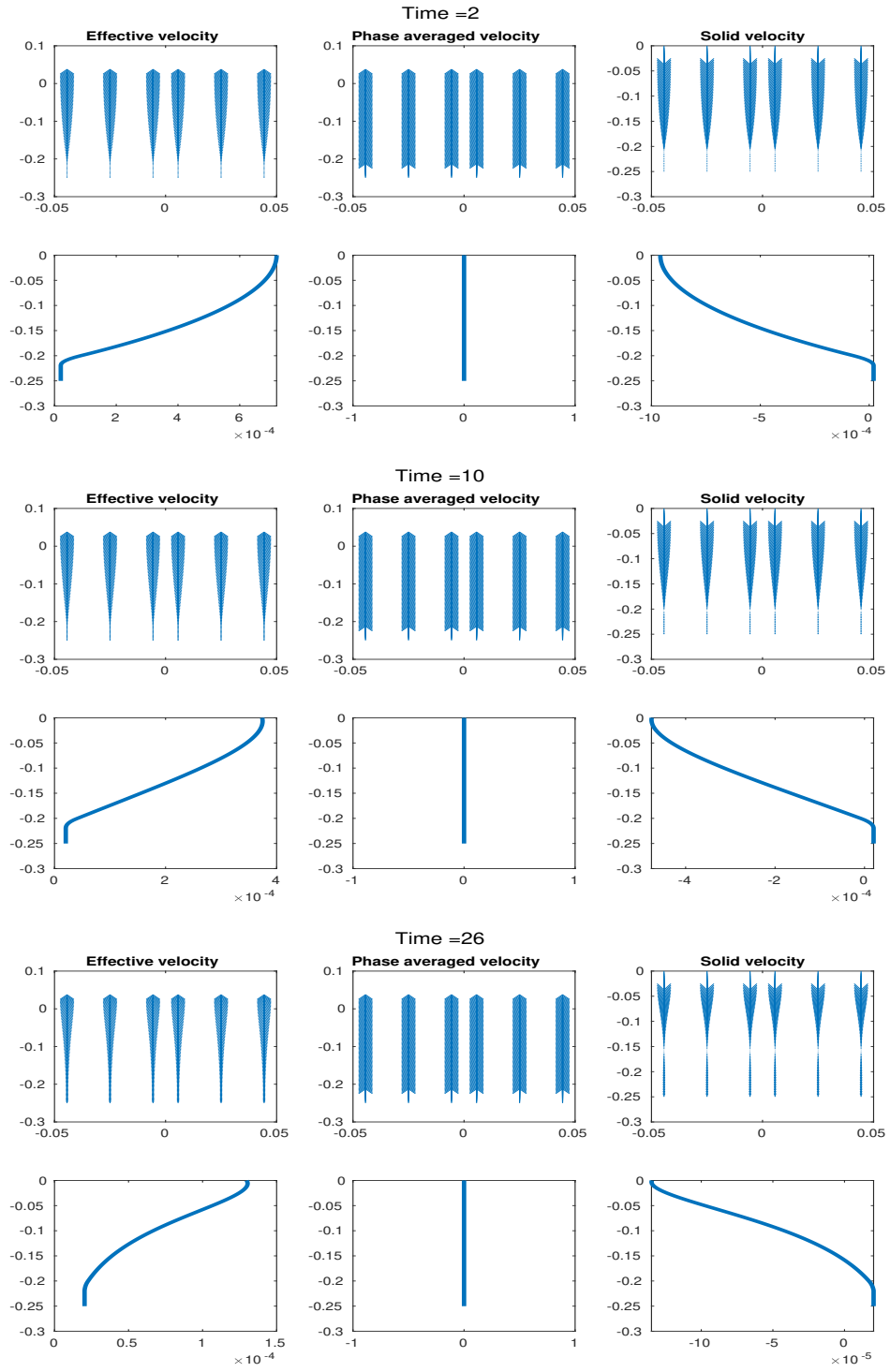


Figure 1.8: Full coupled column testing case $H = 0.25$. Shown at time = 2, 10, 26, 40, and 240. We present porosity, temperature, dimensionless enthalpy, dimensionless mass composition, phase regions and Opx percentage from left to right.



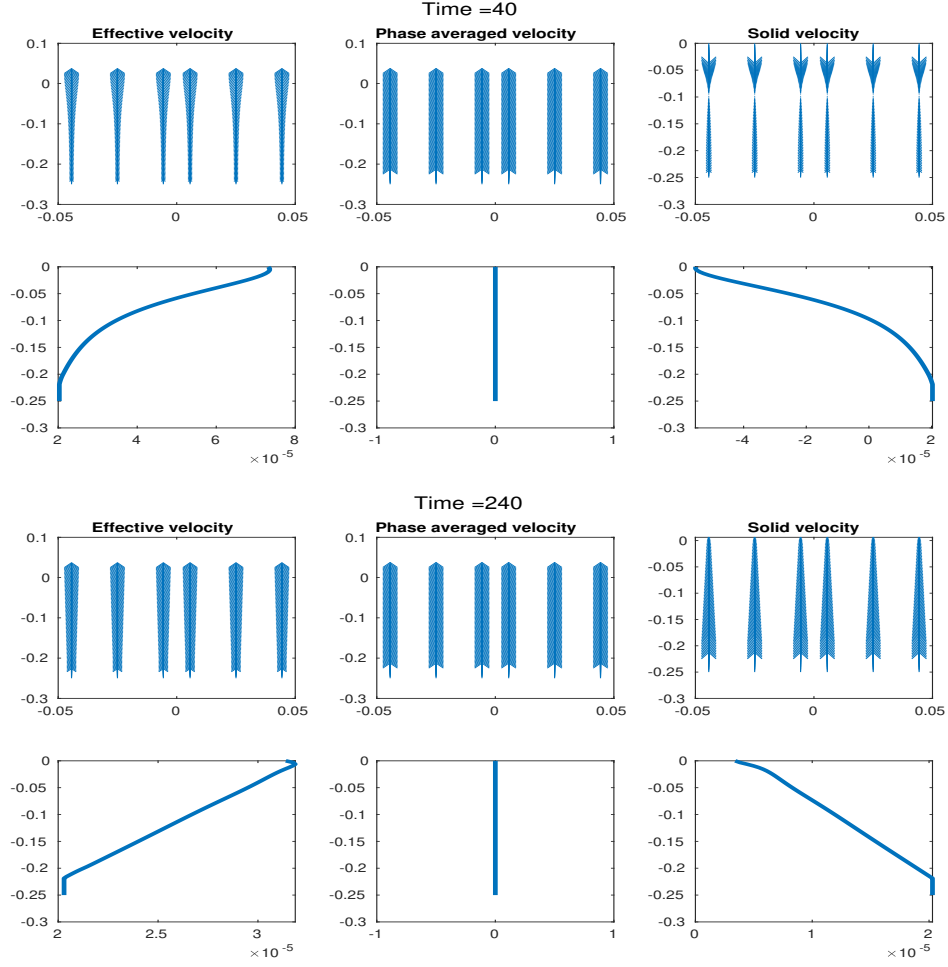


Figure 1.9: Full coupled column testing case $H = 0.25$. Shown at time = 2, 10, 26, 40, and 240. We present effective velocity, phase averaged velocity and solid velocity from left to right.

Chapter 2: Concluding Remarks

2.1 Summary

In this work, we first presented a physics model capturing three key aspects of a two-phase, partially molten mantle system: (1) a degenerate Darcy–Stokes system describing the static mechanical behavior, (2) a nonlinear system governing the transport of chemical species and thermal energy, and (3) a eutectic phase relation characterizing phase transitions. Building upon these governing principles, we developed a locally mass conservative numerical framework that is capable of resolving the coupled multiphysics interactions, with particular emphasis on accurately modeling melt extraction and properly handling degeneracies in regions devoid of melt.

For the degenerate Darcy–Stokes system describing the static mechanics system, we introduced $H(\text{div})$ - and H^1 -conforming mixed finite element spaces for the Darcy and the Stokes subproblems, respectively, based on the mathematical theory developed in Arbogast et al. (2017). Specifically, we employed the first- and second-order direct serendipity spaces designed for the quadrilateral mesh. The $H(\text{div})$ -conforming space relates to the direct serendipity space through the de Rham complex. For the H^1 -conforming component, we proposed a modified Bernardi–Raugel element. We proved the stability and approximation properties of this new element. The resulting saddle point system, arising from the coupling of the degenerate Darcy–Stokes equations, was solved using a modified Uzawa iterative method.

To address the nonlinear transport of chemical species and energy in the presence of a porous media, we adopted and modified the weighted essentially non-oscillatory (WENO) method. In particular, we proposed two alternatives to the classic Jiang–Shu smoothness indicator in order to reduce computational cost (see Huang et al. (2023) and Arbogast et al. (2025)). Furthermore, we introduce a novel multi-level WENO reconstruction scheme (see Arbogast et al. (2024)) that is more versatile

and flexible, allowing the combination of arbitrary stencils of different orders in the reconstruction. This allowed us to be more deal with the complicated multiphysics problems encountered.

Finally, we connected and advanced the multiphysics system by incorporating a eutectic phase module that couples bulk mass composition and thermal energy information with porosity to dynamically capture both the evolution of porosity and phase transitions. This module closes the feedback loop between mechanical, transport, and phase processes, enabling fully coupled simulations of partially molten mantle dynamics.

The computational framework was implemented in C/C++ with PETSc, where, apart from basic linear solvers and standard data management utilities, all components of the numerical algorithms and multiphysics coupling strategies were designed and coded by the author.

2.2 Conclusions

In this dissertation, we developed and analyzed a new Bernardi–Raugel element that achieves the correct rate of convergence on a non-rectangular quadrilateral mesh. We then demonstrated that the $H(\text{div})$ - and H^1 -conforming mixed finite element two-phase Darcy–Stokes solver can seamlessly and robustly handle the degeneracy induced by no-melt regions while maintaining both stability and computational efficiency. Numerical experiments using manufactured pseudo–one-dimensional problems confirmed second-order accuracy in velocity and first-order accuracy in pressure. Furthermore, we showed that the newly developed multilevel weighted essentially non-oscillatory (ML-WENO) reconstruction scheme effectively captures complex transport phenomena and handles diverse boundary conditions with stability and accuracy.

The proposed coupling strategy between the MFEM and FVM solvers proved to be accurate, stable, and computationally efficient, both with and without the in-

volvement of the phase package. The new ML-WENO reconstruction scheme was shown to couple stably and efficiently with the Darcy–Stokes MFEM solver. When the phase package was incorporated into the porosity computation, porosity exhibited strong sensitivity to variations in mass composition and enthalpy, requiring smaller time steps for stability. Nonetheless, the coupled framework accurately resolved complex porosity structures and demonstrated robust performance across all tested scenarios. Boundary conditions were found to play a critical role in ensuring both the physical realism of the simulations and the numerical stability of the solver.

2.3 Future Work

We propose to enhance the stability, versatility, and usability of the framework. This may include incorporating additional stabilization techniques, improving the coupling strategies between multiphysics components, and extending the implementation to general unstructured meshes. Such improvements will enable the framework to address problems involving complex geological geometries. Furthermore, integration and modification with high-performance parallel computing architectures will allow large-scale simulations.

Moreover, the framework opens the possibility of addressing geologically significant problems that remain poorly understood. For example, while melt extraction at mid-ocean ridges has been extensively studied, the style and spatial variability of melt extraction in subduction zones are less constrained. Applying the present framework to such settings may provide new insights into problems like the relationship between melt generation and devolatilization of the slab and melt migration pathways, thereby contributing to a more complete understanding of volatile cycling and magma genesis in subduction settings.

Finally, the generality of the framework lends itself to applications beyond mantle dynamics. With appropriate modifications to the phase relations and constitutive laws, it can be adapted to study a wide range of two-phase partially molten

systems, including problems such as glacial melting, extraterrestrial magmatism, and industrial processes such as eutectic alloy solidification. Such extensions would further demonstrate the flexibility and broad applicability of the numerical methodology developed in this work.

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